Revanth Umakanth

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Summary

Mechanical Engineering student at Vellore Institute of Technology focused on mechatronics, embedded systems, and automation, with hands-on experience integrating sensors, actuators, and controllers into robotic and automated systems. Built a Raspberry Pi-controlled robotic pick-and-place system with machine-vision material classification and ultrasonic height sensing. Industrial experience in production-line automation assessment and Continuous Improvement using Lean Manufacturing and Six Sigma.

Core Competencies: Mechatronics | Embedded Systems | Automation | Robotics | Sensors and Actuators | Raspberry Pi and Arduino | PID Control | Hardware-Software Interfacing | CAD and 3D Printing | Lean Manufacturing | Six Sigma

Education

Vellore Institute of Technology (VIT) — Vellore, Tamil Nadu
B.Tech, Mechanical Engineering | CGPA: 8.04/10 — *2023 - 2027*

Experience

Industrial Engineering Intern — *Jul 2026*
Amara Raja Energy & Mobility Ltd. (AMARON Batteries) — *Tirupati, India*

- *Interned in the CI and Lean department; assessed manual operations for automation potential based on productivity, consistency, and operator workload.*
- *Studied a hydraulic pick-and-place system for heavy industrial batteries and plotted routes for a planned tow-truck battery-transport system.*
- *Conducted time-and-motion and cycle-time analysis (VideoTimerPro), applying Lean and Six Sigma methodologies to identify production bottlenecks.*
- *Proposed process improvements projected to reduce fixed delay time by 16.56 percent and improve full insertion process efficiency by 7.14 percent.*

Engineering Intern — *Jun 2025 - Jul 2025*
Edomotics Smart Systems Pvt. Ltd. — *Hyderabad, India*

- *Worked on mechatronics and home automation systems involving sensor and actuator integration and Control4 platforms, supporting embedded systems development and hardware-software interfacing for automated control applications.*

Projects

Development of an Edge-AI-Based Intelligent Pick-and-Place Robotic System with Material-Based Drop Height Control | *Python, Machine Learning, Edge AI, Image Processing, Raspberry Pi, Dobot Magician Robotic Arm (Capstone Project)*

- *Developed an Edge AI-based pick-and-place robotic system that classifies an object's material (metal, glass, wood, or plastic) and selects a suitable drop height for safe, efficient handling, using a Dobot Magician 4-DOF robotic arm controlled via Raspberry Pi.*
- *Built and tested a basic pick-and-place operation, integrated the Raspberry Pi with the Dobot, and developed robot-control commands through the Raspberry Pi terminal.*
- *Collected and prepared an image dataset across the four material categories, and trained a machine learning model for image-based material classification to enable material-based drop-height selection.*

AI-Assisted Route Optimization for Heavy-Duty Trucks | *Python, Machine Learning, Random Forest, XGBoost, Gemini API, OSMnx, Geospatial Analysis (Problem statement by Volvo India)*

- *Built a machine learning-based route optimization system for heavy-duty trucks, applying data-driven decision-making and engineering problem-solving to reduce fuel consumption, braking events, and carbon emissions on long-haul routes.*
- *Engineered features from GPS and road-network data using OSMnx, and trained and validated Random Forest and XGBoost regression models to evaluate route characteristics and predict optimal routes.*
- *Integrated the Gemini API to generate natural-language route summaries and driver-facing recommendations for AI-assisted route analysis.*

Technical Skills

Mechatronics & Embedded Systems: *Embedded Systems, Mechatronics, Automation, Sensors and Actuators, Raspberry Pi, Arduino, PID Control, Control4*
Robotics: *Robot Programming, Dobot Magician, Collaborative Robots (Cobots), End-Effectors, Automation Integration*
Design & CAD: *SolidWorks, CAD Modeling, 3D Printing, Rapid Prototyping*
Simulation & Analysis: *ANSYS APDL (FEA, meshing, structural analysis)*
Programming, Data & ML: *Python, MATLAB, Excel, Scikit-learn, Random Forest, XGBoost, Gemini API*
Manufacturing & Continuous Improvement: *Lean Manufacturing, Continuous Improvement (CI), Six Sigma, 5S, Time and Motion Study, Cycle Time Analysis*

Certifications

- ***Certificate in Collaborative Robotics in Manufacturing*** *- L&T (Larsen & Toubro)* — *22 May 2025*
- *Completed training in industrial robotics, collaborative robots (cobots), robot programming, automation integration, end-effectors, and industrial safety.*